

Govardhana Kondapaturi

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Work Authorization: F-1 OPT (3-yr US work authorization, no immediate sponsorship required).

SUMMARY

Physical design engineer with ~2 years of graduate research and end-to-end RTL-to-GDS ownership on multi-clock-domain SoCs. Drove a 180K-gate, 12 async-domain chiplet SoC to 513 MHz on Nangate 45nm in Cadence Innovus with structured placement, customized CTS, MCMM signoff in PrimeTime, clean Calibre DRC/LVS and RedHawk-SC IR/EM analysis. Currently taping out an all-digital PLL on SkyWater SKY130 via TinyTapeout SKY26c (Sept 2026); chiplet SoC dissertation submitted to IEEE LASCAS 2027. Comfortable with multi power domain UPF, timing/IR ECOs, formal equivalence in Formality, and TCL/Python automation. M.S. Electrical and Computer Engineering, August 2026.

TECHNICAL SKILLS

- Physical Design & Implementation:** Cadence Innovus, floorplanning, structured placement, macro placement, chip integration, wire editing, power/clock distribution, physical clock design and customized CTS, routing, congestion analysis, timing closure, ECO methodology, multiple power domain implementation (UPF, clock gating, multi-Vt, isolation, level shifters)
- Static Timing & Physical Verification:** Synopsys PrimeTime, MCMM multi-corner setup/hold analysis, SDC constraint development, signoff STA, RedHawk-SC IR-drop and EM analysis, signal EM and noise analysis, Calibre DRC/LVS/PEX/ANTENNA/ERC physical verification signoff, gate-level simulation with SDF annotation
- RTL, Synthesis & Formal:** Verilog, SystemVerilog, Synopsys Design Compiler, DFT-aware synthesis, SpyGlass CDC/RDC and lint, JasperGold, Conformal and Formality equivalence checking
- Programming & Automation:** Python, TCL, Perl, C/C++, STA report parsing, ECO and regression flow automation, Linux

PROJECTS

All-Digital PLL Tapeout on SkyWater SKY130 (TinyTapeout SKY26c Shuttle)

Fresno, USA

Sole Designer (Academic) | Aug 2026 - Sept 2026, Silicon Submitted on Sept 7, 2026

- Designed an all-digital PLL (time-to-digital phase detector, digital PI loop filter, ring-oscillator DCO, programmable feedback divider) in synthesizable Verilog and hardened through OpenLane/OpenROAD on the SkyWater SKY130 open production PDK.
- Closed timing at 40 MHz with sub-10% IR-drop, clean DRC/LVS/ANTENNA via Magic and KLayout on SKY130 standard cells, and submitted GDSII to the ChipFoundry MPW shuttle for real silicon fabrication.

Multi-Chiplet SoC with DVFS, CDC, and Hierarchical Power Management (RTL-to-GDSII)

Fresno, USA

Team Lead (Academic) | Jan 2025 - May 2025

- Owned the full Cadence Innovus flow on a 180K-gate, 12 async-domain SoC, from floorplanning and power planning through CTS, routing, and 513 MHz timing closure with MCMM signoff in PrimeTime.
- Scripted timing and IR-drop ECOs in TCL across corners and reached 100% structural CDC/RDC closure in SpyGlass and JasperGold with clean LEC signoff in Conformal.

ASIC Implementation of 8x8 Systolic Neural Network Accelerator (RTL-to-GDSII)

Fresno, USA

Sole Designer (Academic) | Oct 2024 - Apr 2025

- Delivered an 8x8 systolic MAC accelerator to GDSII in ~64,000 μm^2 with positive setup slack, running Design Compiler and Innovus on Nangate 45 nm with SDC-driven multi-corner analysis.
- Cut dynamic power 38% to 16.8 mW through clock gating, switching-activity optimization, and PPA iterations tracked across Python-parsed PrimeTime reports.

DFT/BIST/ATPG Flow Implementation with Synopsys TetraMAX

Fresno, USA

Sole Designer (Academic) | Aug 2024 - Feb 2025

- Integrated scan chains and BIST (LFSR/MISR) into an FSM controller through DFT-aware synthesis, then generated ATPG patterns in Synopsys TetraMAX to reach 90.93% stuck-at fault coverage.
- Stitched and balanced scan chains with test-mode timing checked in shift and capture, validating the architecture across RTL and gate-level simulation.

EDUCATION

Master of Science in Electrical and Computer Engineering

Fresno, USA

California State University | GPA: 3.3/4.0 | Aug 2024 – August 2026

Bachelor of Technology in Electronics and Communication Engineering

Vijayawada, India

Vasireddy Venkatadri Institute of Technology | GPA: 3.56/4 | Aug 2019 – Aug 2023

EXPERIENCE

Research Assistant – Advanced VLSI Design

California State University, Fresno, USA | Aug 2024 – Aug 2026

- Conducted graduate research with Prof. Aaron Stillmaker across multi-chiplet SoC architectures, UPF-driven low-power multi-voltage implementation, NoC-based physical design, and RTL-to-GDSII signoff on Nangate 45nm (Innovus, PrimeTime, DC, SpyGlass, JasperGold); mentored ~25 graduate students through STA, SDC, and CDC/lint debugging.

PUBLICATIONS

- G. Kondapaturi and A. Stillmaker, "UPF-Driven Multi-Supply Voltage Physical Implementation of a NoC-Based Three-Chiplet System-on-Chip," submitted to IEEE LASCAS 2027.

CERTIFICATIONS

- Synopsys:** [PrimeTime Badge](#), [Fusion Compiler Design Implementation](#). **Cadence:** [RTL-to-GDSII v7.0 \(Advanced\)](#), [SystemVerilog for Design & Verification](#), [STA v3.0](#).